

AI-POWERED COLLEGE ASSISTANT USING LANGCHAIN TOOL CALLING AGENT

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ABSTRACT

The AI-Powered College Assistant is an intelligent application developed using Python, LangChain, and Ollama. The system automatically understands student queries and invokes the appropriate tool to provide accurate responses. The assistant performs various academic-related tasks such as attendance calculation, result analysis, fee balance calculation, library fine calculation, hostel fee calculation, and student information retrieval. LangChain Tool Calling Agent enables automatic tool selection and execution based on user requests.

OBJECTIVES

The main objectives of this project are:

1. To develop an AI-powered assistant for college-related activities.
 2. To implement tool calling using LangChain.
 3. To automate attendance and result calculations.
 4. To calculate fee balances and library fines.
 5. To demonstrate multi-tool execution using an AI agent.
 6. To retrieve student information using student IDs.
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TECHNOLOGIES USED

- Python
 - LangChain
 - Ollama
 - ChatPromptTemplate
 - AgentExecutor
 - create_tool_calling_agent()
 - Tool Decorators (@tool)
 - PyCharm IDE
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SYSTEM ARCHITECTURE

User Query



LangChain Tool Calling Agent



Tool Selection



Appropriate Tool Execution



Response Generation



Final Output

The agent analyzes the user query and automatically invokes the relevant tool based on the request.

IMPLEMENTED TOOLS

1. Attendance Calculator

Inputs

- Total Classes
- Attended Classes

Outputs

- Attendance Percentage
- Exam Eligibility Status

Rule

Attendance $\geq 75\%$ $\rightarrow$ Eligible for Exam

Attendance $< 75\%$ $\rightarrow$ Not Eligible for Exam

2. Result Calculator

Inputs

- Marks of 5 Subjects

Outputs

- Average Marks
- Grade
- Pass/Fail Status

Grading Criteria

Average Marks Grade

≥ 90	A
75–89	B
60–74	C
< 60	D

Pass if Average ≥ 50

3. Fee Balance Calculator

Inputs

- Total Course Fee
- Amount Paid

Output

- Pending Fee Amount

Formula:

$$\text{Pending Fee} = \text{Total Course Fee} - \text{Amount Paid}$$

4. Library Fine Calculator

Input

- Delayed Days

Output

- Fine Amount

Formula:

$$\text{Fine} = ₹5 \times \text{Delayed Days}$$

5. Hostel Fee Calculator

Inputs

- Monthly Hostel Fee
- Number of Months Stayed

Output

- Total Hostel Fee

Formula:

Hostel Fee = Monthly Fee $\times$ Number of Months

6. Student Information Tool

Input

- Student ID

Output

- Student Details

Information retrieved from a Python dictionary.

IMPLEMENTATION

The project uses LangChain's Tool Calling Agent to determine which tool should be invoked based on the user's request. Each functionality is implemented using the `@tool` decorator. The `create_tool_calling_agent()` function creates the intelligent agent, while `AgentExecutor` manages execution and verbose output.

TEST CASES AND OUTPUTS

Test Case 1

Input

I attended 72 classes out of 90. Am I eligible for exams?

Output

Attendance Percentage: 80.00%

Exam Eligibility Status: Eligible for Exam

```
QUERY: I attended 72 classes out of 90. Am I eligible for exams?
=====

> Entering new AgentExecutor chain...

Invoking: 'attendance_calculator' with {'attended_classes': 72, 'total_classes': 90}

Attendance Percentage: 80.00%
Exam Eligibility Status: Eligible for ExamBased on your attendance of 72 out of 90 classes, your attendance percentage is 80.00%. You are eligible to write the exams.

> Finished chain.
Based on your attendance of 72 out of 90 classes, your attendance percentage is 80.00%. You are eligible to write the exams.
\n=====
```

Test Case 2

Input

My marks are 95, 90, 88, 91 and 87. What is my grade?

Output

Average Marks: 90.20

Grade: A

Status: Pass

```
QUERY: My marks are 95, 90, 88, 91 and 87. What is my grade?
=====

> Entering new AgentExecutor chain...

Invoking: 'result_calculator' with {'mark1': 95, 'mark2': 90, 'mark3': 88, 'mark4': 91, 'mark5': 87}

Average Marks: 90.20
Grade: A\nStatus: PassBased on the marks you provided, your average is 90.20 and you have been graded as 'A'. Congratulations, you have passed!

> Finished chain.
Based on the marks you provided, your average is 90.20 and you have been graded as 'A'. Congratulations, you have passed!
\n=====
```

Test Case 3

Input

My course fee is 50000 and I have paid 35000.

Output

Pending Fee Amount: ₹15000

```
QUERY: My course fee is 50000 and I have paid 35000. How much fee is pending?
=====

> Entering new AgentExecutor chain...

Invoking: `fee_balance_calculator` with `{'total_course_fee': 50000, 'amount_paid': 35000}`

Pending Fee Amount: ₹15000.00The pending fee amount for your course is ₹15000.00.

> Finished chain.
The pending fee amount for your course is ₹15000.00.
```

Test Case 4

Input

I returned a library book 8 days late.

Output

Fine Amount: ₹40

```
QUERY: I returned a library book 8 days late. What is the fine amount?
=====

> Entering new AgentExecutor chain...

Invoking: `library_fine_calculator` with `{'delayed_days': 8}`

Fine Amount: ₹40The library fine for returning the book 8 days late is ₹40.

> Finished chain.
The library fine for returning the book 8 days late is ₹40.
```

Test Case 5

Input

Hostel fee is 6000 per month and I stayed for 5 months.

Output

Total Hostel Fee: ₹30000

```
QUERY: Hostel fee is 6000 per month and I stayed for 5 months. Calculate my hostel fee.
=====

> Entering new AgentExecutor chain...

Invoking: `hostel_fee_calculator` with `{'monthly_hostel_fee': 6000, 'months_stayed': 5}`

Total Hostel Fee: ₹30000.00Your total hostel fee for 5 months is ₹30,000.00.

> Finished chain.
Your total hostel fee for 5 months is ₹30,000.00.
```

Multi-Tool Challenge

Input

I attended 80 classes out of 100.

My marks are 90, 85, 88, 92 and 95.

My course fee is 60000 and I paid 45000.

Output

Attendance Percentage: 80.00%

Exam Eligibility Status: Eligible for Exam

Average Marks: 90.00

Grade: A

Status: Pass

Pending Fee Amount: ₹15000

```
QUERY: I attended 80 classes out of 100.
My marks are 90, 85, 88, 92 and 95.
My course fee is 60000 and I paid 45000.

Provide:
1. Attendance Status
2. Grade
3. Pending Fee
=====

> Entering new AgentExecutor chain...

Invoking: `attendance_calculator` with `{'attended_classes': 80, 'total_classes': 100}`

Attendance Percentage: 80.00%
Exam Eligibility Status: Eligible for Exam
Invoking: `result_calculator` with `{'mark3': 88, 'mark4': 92, 'mark5': 95, 'mark1': 90, 'mark2': 85}`

Average Marks: 90.00
Grade: A\nStatus: Pass
Invoking: `fee_balance_calculator` with `{'total_course_fee': 60000, 'amount_paid': 45000}`

Pending Fee Amount: ₹15000.00Based on the provided information:
```

BONUS FEATURE

Student Information Retrieval

Input

Show details for student ID 101

Output

Name: Priya

Department: CSE

Year: 3

```
QUERY: Show details for student ID 101
=====

> Entering new AgentExecutor chain...

Invoking: 'student_information' with `{'student_id': 101}`

{'name': 'Priya', 'department': 'CSE', 'year': 3}The details for student ID 101 are as follows:
- Name: Priya
- Department: CSE
- Year: 3rd Year

> Finished chain.
The details for student ID 101 are as follows:
- Name: Priya
- Department: CSE
- Year: 3rd Year
```

RESULTS

The developed AI-powered assistant successfully:

- Calculated attendance percentages.
- Determined exam eligibility.
- Calculated grades and pass/fail status.
- Computed pending fees.
- Calculated library fines.
- Calculated hostel fees.
- Retrieved student information.
- Executed multiple tools automatically for complex queries.

CONCLUSION

The AI-Powered College Assistant was successfully developed using LangChain Tool Calling Agent and Ollama. The system automatically identifies user requests and invokes appropriate tools to generate accurate responses. The project demonstrates the practical implementation of tool calling agents and highlights the capabilities of modern AI frameworks in educational applications.

FUTURE ENHANCEMENTS

1. Integration with a real student database.
2. Web-based user interface.
3. Attendance prediction using Machine Learning.
4. Automatic timetable management.
5. College notification and event management.
6. Integration with college ERP systems.